Blockchain

Blockchain Management

v1.0.0

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Transaction Types



Transaction

- From Address
- To Address
- Amount
- Digital Signature

Ethereum Transaction

- Nonce (number used only once)
- GasPrice (how much you’re willing to pay for gas)
- GasLimit (how much gas you’re willing to consume)
- Recipient
- Value (can be zero – if it’s a contract interaction, for example)
- Data (can be empty – if it’s a value transfer, for example)
- Signature

The multi-signature transaction

- Available in Bitcoin, but not in Ethereum (without smart contract)
- N signatures required to perform interaction
- M of N signatures possible
- Many uses, such as wallets and forms foundation of layer 2 scaling in BTC

Transaction to Block



Use Cases



Source: Deloitte

Blockchain Use Cases

- Store of Value
- Transfer of Value
- Identity
- Asset Tracking (property, land, warranties)
- Track and Trace
- Trade Finance
- Counterfeit Protection
- Decentralised Finance

Store of Value

- Cryptocurrencies as digital gold
- Does not require medium of exchange properties
- Secure, transferrable, digital, global

Transfer of Value

- Digital transference of value via transaction
- No third party
- Global, digital, instant?

Asset Tracking

- Tokenisation of real world assets
  - Property
  - Land
  - Shares
  - Warranty
- Strict ownership of asset by private key
- Data migration issues
- Still need for real world entities
- Challenges associating real-world entity with digital asset

Track and Trace

- Blockchains not useful for collation of tracking and tracing data
- Typically tracking and tracing data provided to single entity
- Data monetisation shows promise
- Some benefits around multiple sources writing to common platform

Trade Finance

- Up to 50% of the cost of a shipping container is its paperwork
- A blockchain can act as the single source of truth
- Letters of credit – prevention of financing fraud
- Large ecosystem challenge, potentially a digitisation challenge, rather than blockchain challenge without sufficient network effects

Counterfeit Prevention

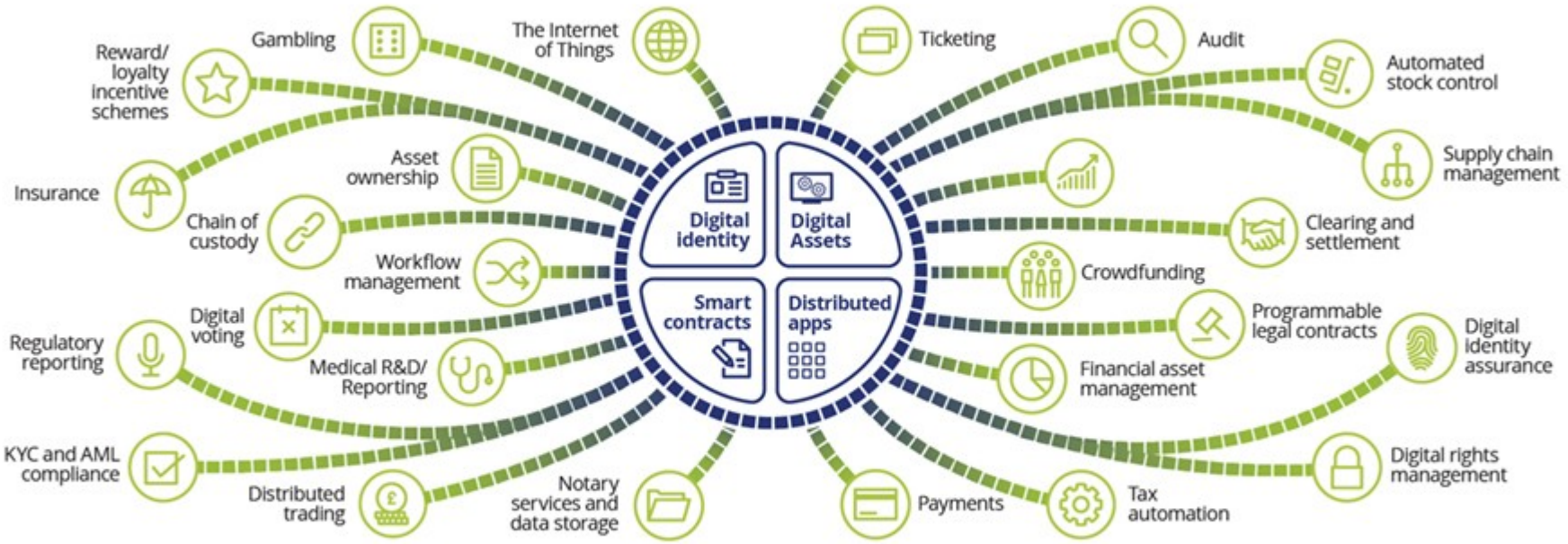
- Determination of chain of custody from manufacturer to purchaser
- Can only sell one counterfeit example
- Tokenisation of luxury good, loyalty schemes
- Blockchain?

Decentralised Finance

- Collaterlised Loans (DAI)
- Flash Loans
- Liquidity Providers
- Oracles
- Decentralised vs. Centralised exchanges

What can you do with a blockchain?

KYC – Know Your Customer
AML – Anti-Money Laundering



Deloitte.

www.deloitte.co.uk/blockchain

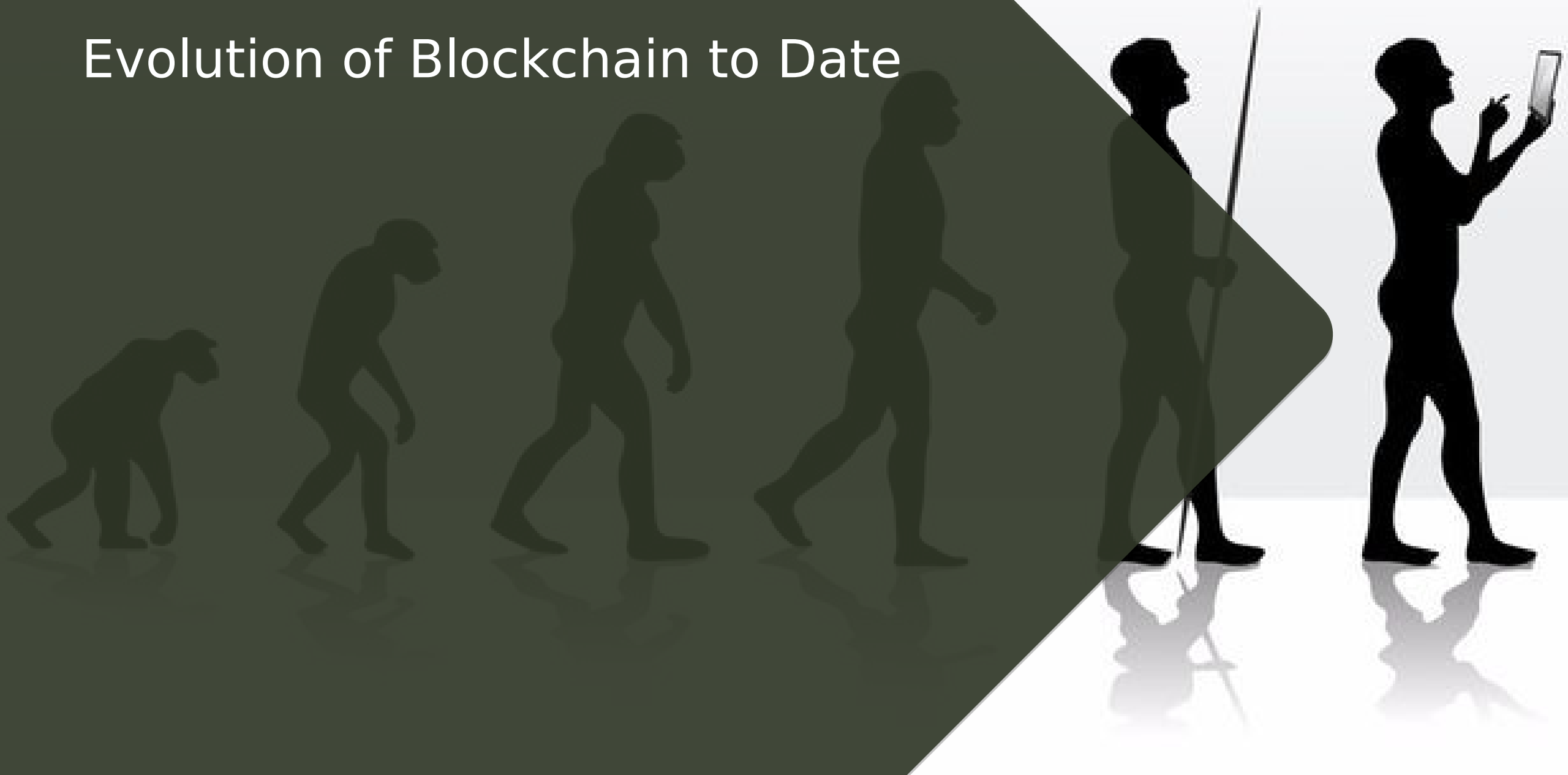
Blockchain Use Cases

- Other than its poor Medium of Exchange properties, bitcoin can also be used as the immutable, cryptographically signed and secured platform for a number of other uses
- Currently digital identities are owned by corporations (Google, Facebook, Twitter, Yahoo, etc.), rather than the individuals. How can blockchain provide an alternative to this?

Identity

- PKI at its core, identity backed by ownership of a private key
- Identity owner pattern (uPort)
- Key recovery and security
- Encryption key vs. Identity key (symmetric vs asymmetric encryption)
- Biometrics are username, not password
- Attestations
- Permissioned Sharing

Evolution of Blockchain to Date



Evolution of Blockchains

- Bitcoin
  - First decentralised cryptocurrency
  - Has limited scripting capabilities (to provide extensibility)
- Ethereum
  - Turing-complete smart contracts
  - Platforms can run directly on-chain

- Current focus is on Scaling and Privacy

Layer 2 Networks

- Lightning Network
- Raiden Network

- Use the blockchain as a settlement layer
- Uses payment channels
- A payment channel requires 2 transactions on-chain (open and close)

Payment Networks

- If Bob has a channel with Alice
- And Bob has a channel with Carol
- Alice can send payments to Carol, through Bob, with the participants updating their channel totals through a number of transaction swaps
- Performed by hashing secrets (Alice demands the secret off Bob, and Bob obtains it from Carol for the payment amount)
- To ensure Carol doesn’t just take Bob’s money, all transactions are hash time-locked transactions

- https://bitcoinmagazine.com/articles/understanding-the-lightning-network-part-building-a-bidirectional-payment-channel-1464710791/

Token Issuance (ICOs)

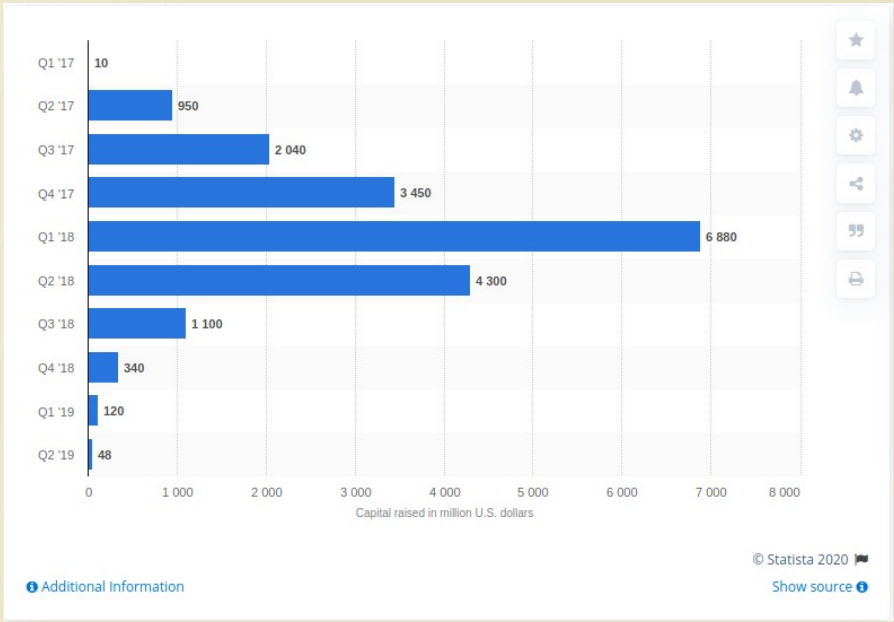
- ERC20 Token smart contract standard
- https://theethereum.wiki/w/index.php/ERC20_Token_Standard

```
// https://github.com/ethereum/EIPs/issues/20
 2 contract ERC20 {
 3     function totalSupply() constant returns (uint totalSupply);
 4     function balanceOf(address _owner) constant returns (uint balance);
 5     function transfer(address _to, uint _value) returns (bool success);
 6     function transferFrom(address _from, address _to, uint _value) returns (bool success);
 7     function approve(address _spender, uint _value) returns (bool success);
 8     function allowance(address _owner, address _spender) constant returns (uint remaining);
 9     event Transfer(address indexed _from, address indexed _to, uint _value);
10     event Approval(address indexed _owner, address indexed _spender, uint _value);
11 }
```


DID SOMEONE SAY ICO?

ICOs

- Issue a token for ether, with the rate set by the issuers
- The token will nominally be required for the platform
- The issuers can potentially create more tokens
- The token is not based on utility, and is the only code required
- ~~$2.4B raised in 2017~~
  - EOS: 4.2B
  - Gram: 1.7B
  - Petro: 735M
  - Filecoin: 257M
  - Ethereum: 5.2M



https://www.statista.com/statistics/804748/worldwide-amount-crytocurrency-ico-projects/

Scaling and Privacy

- Scaling
  - On-chain scaling will never work without centralisation occurring
  - Layer 2 networks required to provide required transaction throughput
  - Layer 3 network papers being released (lightning on lightning)
- Privacy
  - All data on-chain is visible to all participants
  - Data on-chain must either be encrypted or provide an anchor to off-chain data
  - Lightning transactions are private, but opening and closing states are public
  - Zcash, Monero provide private transactions via different mechanisms

Sharding

- Sharding is splitting the chain into multiple components, with each component only having to keep track of its own data
- Communication with other shards will be via asynchronous transactions
- Sharded chains do not have to be compatible
- Sharded chains are likely to have varying levels of security (particularly compared to the main chain)

Distributed Computing

- Paid to provide CPU cycles
- Paid to provide output to signed code
- Requires payment channels rather than on-chain transactions

Mixers and Tumblers

- Send your coins to a service
- Your coins get mixed with other coins and you get someone else’s coins
- Breaks the chain of custody of coins
- Requires trust of mixing service
- Some currencies (Dash) mix by default
- Mixing creates tainted coins (fungibility concern)
- Risk of a smaller than expected anonymity set

Ring Signatures

- Anyone in a group of keys can sign the transaction
- Computationally infeasible to determine who actually signed the transaction
- Invented by Ron Rivest, Adi Shamir, and Yael Tauman in 2001.
- Used by Monero to provide privacy

Zero Knowledge Proof

- Can be used to reveal a proof of a secret, without revealing the secret
- New crypto, “"The Knowledge Complexity of Interactive Proof-Systems" written in 1985 by Shafi Goldwasser, Silvio Micali, Charles Rackhoff
- Used in Zcash to prove the validity of a transaction without revealing the sender, recipient or amount
- Confidential transactions not mandatory in Zcash, and a confidential transaction takes several minutes to create

Homomorphic Encryption

- Enables computation on encrypted data
- For example, adding an encrypted number to another encrypted number to provide a new encrypted number that is a sum of the inputs
- Computationally expensive, but getting faster slowly

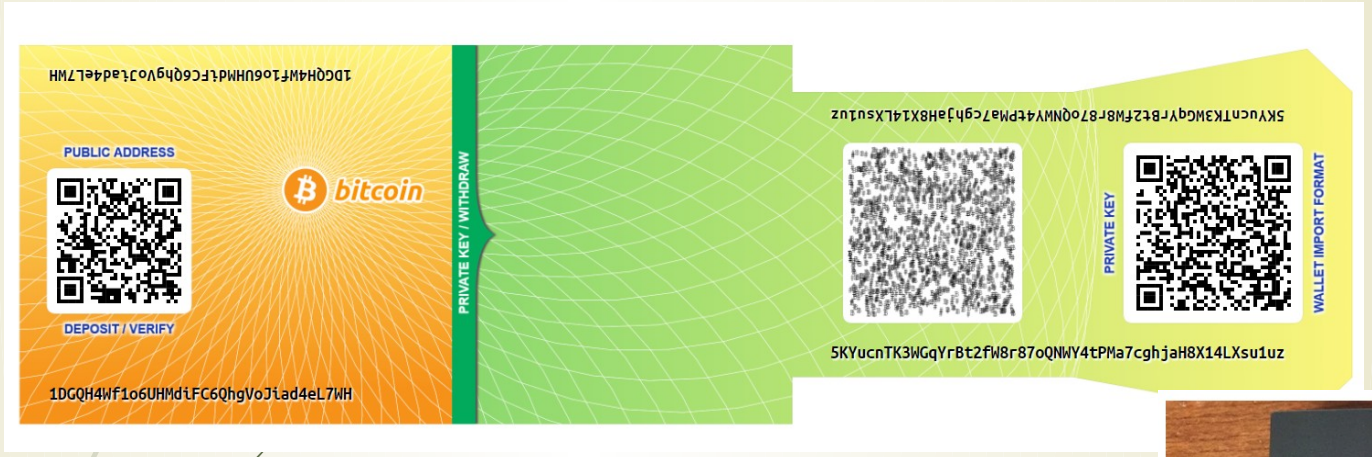
Storing and Using Cryptocurrencies



Storing and Using Cryptocurrencies

- Exchange (trusted third party: Coinbase, XAPO, Bitfinex)
- Full Node
- Software Wallet (mobile phone, PC)
- Paper Wallet
- Hardware Wallet

Paper & Hardware Wallets





Blockchain Ledgers



Private Blockchains

- Ethereum
  - Uses public Ethereum client (Geth, Parity) or private (Quorum)
  - Gas not a concern, no value in private Ether
  - Can use different consensus mechanisms (PoA, PBFT)
- Hyperledger
  - Created by IBM, now open source (Hyperledger Foundation)
  - private only, no public implementation
  - contains centralised components (ordering service)
  - has private channels & high transaction throughput (700 tps)

Mining



Mining

- Mining is the validation of blocks of transactions
- Proof of Work mining involves hashing the block with a nonce to obtain a hash below a certain value
- Proof of Work mining makes generating a valid block difficult to prevent the network being flooded with invalid blocks (PBFT solution)
- Attempts underway in Ethereum to move to Proof of Stake

History of Mining

- Initially mining was performed on desktop/laptop CPUs
- Then it moved to GPUs
- Then ASICs – specific chips whose only purpose is mining

- No longer efficient to mine without ASICs, and cheap/free electricity is essential to maintain profitability
- Note that Ethereum uses ASIC-resistant hashing algorithm (ethhash, variation of Dagger-Hashimoto), so GPUs must be used

Mining Centralisation

- Main mining hardware companies
  - Bitmain
  - Bitfury (don’t sell their hardware)
  - GMO (coming – but might be private operation)

- Additionally 60% of bitcoin mining operations are based in China
- Possibility that 10 individuals control 60% of mining power

Centralisation vs Decentralisation



Centralised Systems

- Excellent potential control and governance
- Rigorous onboarding (KYC)
- Trusted Third Parties
- Scale and Privacy
- Honeypots (2017 Equifax, 2006 Israeli Theft)
- Centralisation of Biometric Data ongoing (Aadhaar – 1bn biometric identities)

Decentralised Systems

- Poor governance by design
- Limited user tracking
- No central data source
- No trusted third parties
- Limited scale
- Censorship resistance

Distributed Consensus



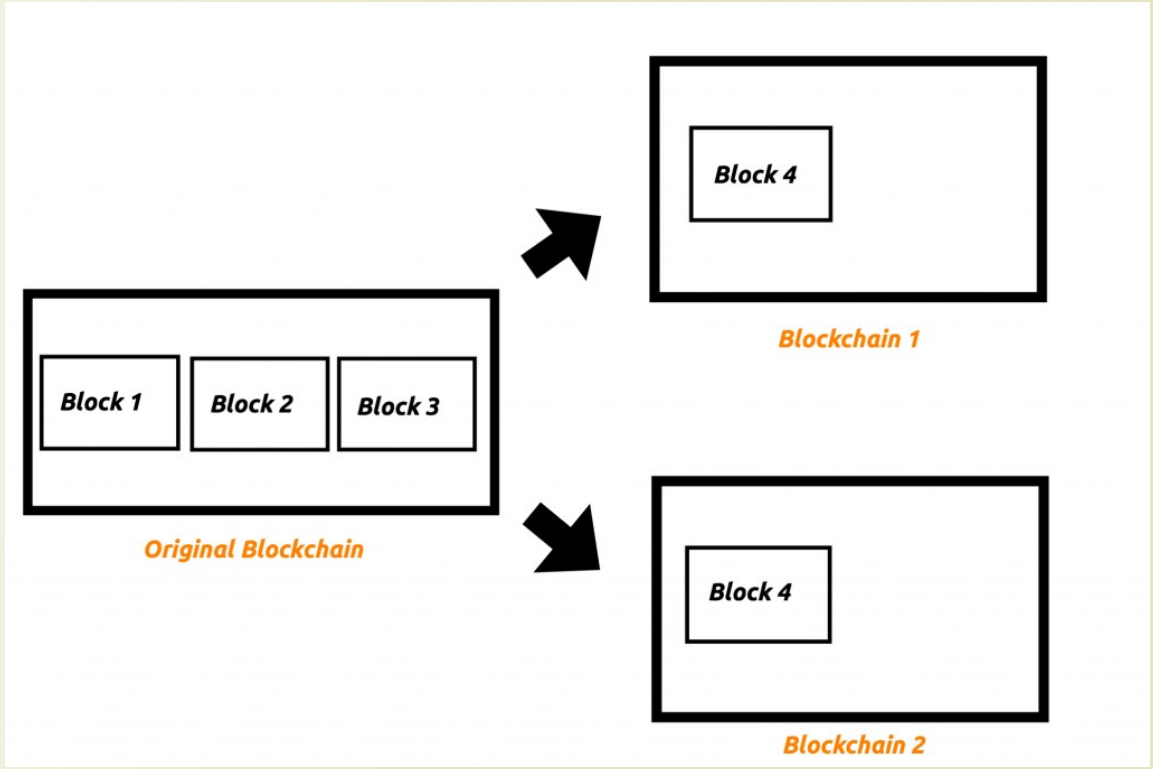
Governance Issues

- Do miners control bitcoin?
- Segwit2X was an attempt by corporations and miners to control bitcoin
- Segwit2X was called off by these same individuals
- There are a number of power structures in Bitcoin
  - Miners
  - Users (light wallets)
  - Full Nodes
  - Developers
  - Hodlers

- Lack of governance and control is a benefit for censorship resistance

Distributed Consensus and Autonomy

- Upgrading blockchain functionality while maintaining backwards compatibility with old clients is performed via a **soft fork**
- Upgrading blockchain functionality and removing backwards compatibility (forcing all clients to upgrade to remain on the valid chain) is called a **hard fork**
- If the network undergoes a hard fork, any clients that have not upgraded will fork to another chain



Examples of Hard Forks

- Ethereum Classic created when Ethereum hard forked to return the stolen DAO funds
- Bitcoin Cash forked from Bitcoin to provide 8MB blocks without segwit
- Bitcoin Gold forked from Bitcoin to provide its creators with 100,000 bitcoin gold (+ Segwit, GPU mining and replay protection)

- Replay Protection is removing the ability for a valid transaction on one fork to be replayed on another fork

- Note: exchanges have fiduciary duty to provide forked coins

Sybil Attacks

- “Proof-of-work is essentially one-CPU-one-vote.”
- Without proof of work, cost of creating a user is free (and there is no central authority to validate uniqueness)
- Ability for single individual to create hundreds or thousands of valid appearing addresses or services easily and cheaply
- Implications for governance

User Activated Soft Fork

- Segregated Witness (BIP141) was an attempted soft fork upgrade to the bitcoin network
- Required 95% miner signalling for 2 weeks by Nov 15th 2017
- Achieved no more than 45% miner agreement (via block signalling)
- UASF (BIP148) was a reaction to the blocking of segwit by miners
- After August 1, any blocks not signalling Segwit would be rejected by nodes running BIP148, creating a fork in the network
- BIP91 created to activate segwit with 80% consensus for 2 days
- Segwit2X agreement gave network segwit and UASF not needed
- Suspicion still that Segwit2X was a reaction to BIP148

Smart Contracts



Smart Contracts

- A contract enforced by code
- Code runs on blockchain, so inherits immutable trust
- Much buzzword, such wow
- Bitcoin scripting is technically smart contract
- Turing-complete smart contracts
  - Provide limitless functionality, all running on chain
  - Provide a limitless surface area for attacks
  - Provide a simple prototyping mechanism for on-chain interactions
  - Are more powerful than bitcoin scripting

Example smart contract

```
contract simpleDNS {
    struct Record {
        address owner;
        string ipAddr;
    }
   mapping (string => Record) records;
   function addDomain(string domain, string ipAddr) {
        if records[domain].owner != msg.sender) { return; }
        records[_domain] = Record(msg.sender, ipAddr);
    }
    function getDomain(string domain) constant returns(string) {
        return records[domain].ipAddr;
    }
    function transfer(string domain, address toAddr) {
        if (records[domain].owner != msg.sender) { throw; }
        records[domain].owner = toAddr;
    }
}
```

Conclusion



Conclusion

- A multisig transaction is a very powerful smart contract
- There are many potential use cases – but Store of Value is primary one in use
- Payment channels are a powerful scaling mechanism, but still require too much on chain interaction
- ICOs are a killer app, new form of VC with a large side-order of scam
- Privacy mechanisms exist, but are complex and not yet in common operation
- Distributed consensus being hard is not a bad thing
- Use a hardware wallet